



BITS, PILANI – K. K. BIRLA GOA CAMPUS

Database Systems (CSF212)

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Assertions

Assertions

- An *assertion* is a predicate expressing a condition that we wish the database always to satisfy.
- An assertion in SQL takes the form
create assertion <assertion-name> **check** <predicate>
- When an assertion is made, the system tests it for validity, and tests it again on every update that may violate the assertion.
 - This testing may introduce a significant amount of overhead; hence assertions should be used with great care.
- Asserting $\text{for all } X, P(X)$ is achieved in a round-about fashion using $\text{not exists } X \text{ such that not } P(X)$

NOTE

- Inside the NOT EXISTS clause specify a query that violates the required condition.
- Inner query result must be empty. If the query result is not empty, the assertion has been violated.

Assertion Example

Consider the following schema:

Loan		
<u>Loan Number</u>	Branch_Name	Amount

Depositor	
<u>Account Number</u>	<u>Customer Name</u>

Account		
<u>Account Number</u>	Branch_Name	Balance

Borrower	
<u>Customer Name</u>	<u>Loan Number</u>

Branch		
<u>Branch Name</u>	Branch_City	Assets

Customer		
<u>Customer Name</u>	Customer_Street	Customer_City

The sum of all loan amounts for each branch must be less than the sum of all account balances at the branch.

create assertion *sum-constraint* check

(not exists (select * from *branch*

where (select sum(*amount*) from *loan,branch*
where *loan.branch-name* =
branch.branch-name)

>= (select sum(*balance*) from *account,branch*
where *account.branch-name* =
branch.branch-name)))

Assertion Example

Consider the following schema:

Loan		
<u>Loan Number</u>	Branch_Name	Amount

Depositor	
<u>Account Number</u>	<u>Customer Name</u>

Account		
<u>Account Number</u>	Branch_Name	Balance

Borrower	
<u>Customer Name</u>	<u>Loan Number</u>

Branch		
<u>Branch Name</u>	Branch_City	Assets

Customer		
<u>Customer Name</u>	Customer_Street	Customer_City

Every loan has at least one borrower who maintains an account with a minimum balance of \$1000.00

create assertion *balance-constraint* check

(not exists (

select * from *loan*

where not exists (

select *

from *borrower*, *depositor*, *account*, *loan*

where *loan*.*loan-number* = *borrower*.*loan-number*

and *borrower*.*customer-name* = *depositor*.*customer-name*

and *depositor*.*account-number* = *account*.*account-number*

and *account*.*balance* >= 1000)))

Assertion Example

The salary of the employee must not be greater than the salary of the manager of the department for which the employee work for.

```
CREATE ASSERTION EMP_SAL_CONSTRAINT  
CHECK (NOT EXISTS  
      (SELECT *  
        FROM EMPLOYEE E, EMPLOYEE M, DEPARTMENT D  
        WHERE E.SALARY > M.SALARY AND E.DNO=D.NUMBER AND D.MGRSSN=M.SSN) )
```

Drop an assertion

To drop an assertion, you can use the following statement:

- **DROP ASSERTION `assertion_name`;**

Assertions are useful for enforcing data integrity and ensuring that the data in the database meets certain conditions. They can be used to enforce business rules or to ensure the consistency of the data.